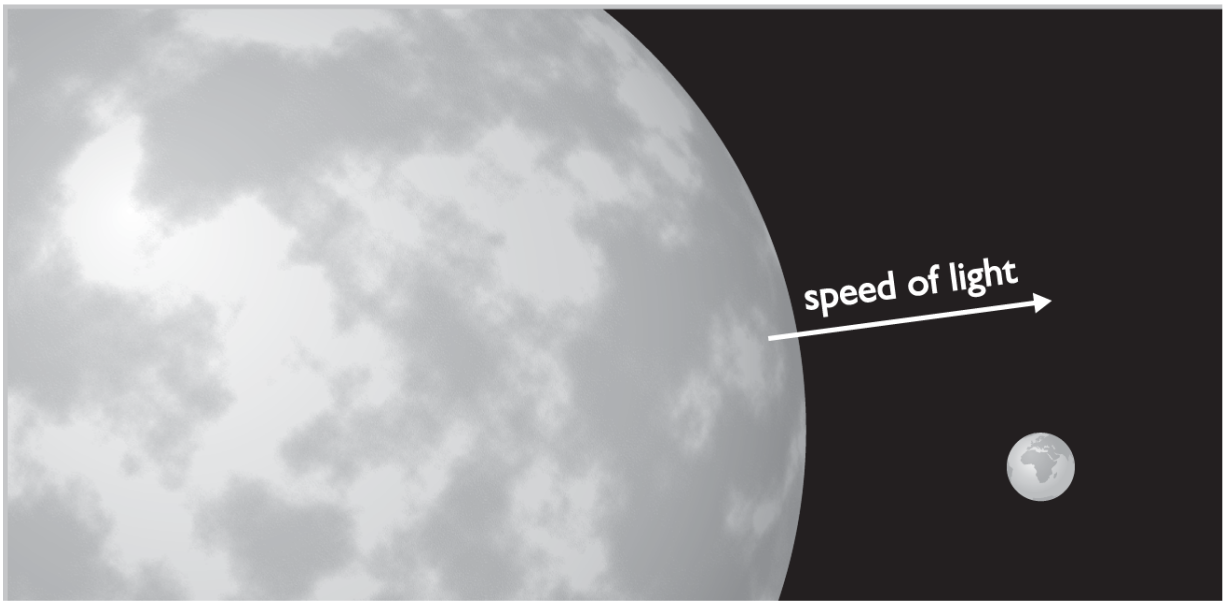


Figure 18: The bubble of true vacuum. If a vacuum decay event happens at one place in the cosmos, it causes a bubble to expand outward at the speed of light, destroying everything in its path.

KICKING THE HORNET'S NEST

You shouldn't worry about vacuum decay. Really. For several reasons. There are the obvious ones, of course: there's no way to stop it if it's happening; and you can't know it's about to; and it's not like it would hurt; and no one would be around to miss you anyway, so what's the point of worrying about it? You're better off double-checking your smoke alarm batteries and, I don't know, lobbying to close down coal power plants or something. But if for some reason that isn't sufficiently reassuring, I can also say with a reasonable degree of certainty that vacuum decay is extremely unlikely to happen—at least, anytime in the next many many many trillions of years.

There are a few ways vacuum decay could, in theory, occur. The most straightforward is some kind of high-energy event. Think of it as the equivalent of an earthquake, knocking the pebble out of its divot to send it plummeting to the valley floor. Fortunately, the "earthquake" in this case would have to be really unfathomably powerful. The best estimates we have suggest that the event would have to be much more energetic than the most devastating explosions we've witnessed in the cosmos, and certainly many orders of magnitude stronger than anything we could possibly do with a human-built machine like the Large Hadron Collider. If we're ever worried